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SUPERSEDING
Navy 52C22
1 June 1944

MILITARY SPECIFICATION

CEMENT, ADHESIVE, INSULATION, FIBROUS

(All interested bureaus of the Navy Department have concurred in the use of this specification for purchase.)

1. SCOPE AND CLASSIFICATION

1.1 Scope.- Fibrous insulation adhesive cement is a cement for securing woven asbestos cloth to insulating material employed on piping or other applications for Naval vessels.

1.2 Type.- Fibrous insulation adhesive cement shall be of but one type as hereinafter specified.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 Specifications.- The following specifications, of the issue in effect on date of invitations for bids, form a part of this specification:

Federal Specifications
~~LLL-B-631 - Boxes, Fiber, Corrugated (For Domestic Shipment).~~
LLL-B-636 - Boxes, Fiber, Solid (For Domestic Shipment).
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed.
JAN-B-107 - Boxes, Wood, Wirebound (Overseas Type).
JAN-P-124 - Packaging and Packing for Overseas Shipment - Containers (Cans, Pails and Drums), Metal (For Other Than Subsistence Items).

Military Specification
MIL-C-1207 - Cloth and Strands, Asbestos.

Navy Department Specifications
General Specifications for Inspection of Material.
32014 - Cement, Insulation, High-Temperature.

(Army.- Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(Navy.- Copies of Federal, Military and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Center, Norfolk 11, Va. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(Air Force.- Copies of Federal and Military specifications may be obtained upon application to the Commanding General, Air Materiel Command, Wright-Patterson Air Force Base, Dayton, Ohio. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(Marine Corps.- Copies of Federal and Military specifications may be obtained upon application to the Quartermaster General, Headquarters U.S. Marine Corps, Navy Department, Washington 25, D. C. or the Depot Quartermaster, Marine Corps Depot of Supplies, 1100 South Broad Street, Philadelphia 46, Pa. Both the title and identifying number or symbol should be stipulated when requesting copies.)

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2.2 Other publications.- The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

Bureau of Supplies and Accounts Publication
Navy Shipment Marking Handbook.

(Copies of Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Center, Norfolk 11, Va.)

3. REQUIREMENTS

3.1 The material shall be free from all ingredients which may effect its serviceability or have a deleterious effect upon the metal or insulation to which the cement is applied.

3.2 The cement, in the condition in which delivered, shall be suitable and effective for the purpose specified hereinafter, without heating or addition of other ingredients; except that it may be furnished in unmixed form in unit containers, to be mixed just prior to use.

3.3 The cement shall not deteriorate when enclosed for an indefinite length of time in airtight metal containers.

3.4 The cement shall be suitable for securing woven asbestos cloth to insulating compound employed on piping or in other applications.

3.5 The adhesive strength of the cement shall be such that, when tested by the method described in 4.4.1, a separation of the applied cloth from the insulating compound will result in rupture of the insulating compound and not of the cement.

3.6 When tested by the method described in 4.4.2, the cement shall not give off objectionable or irritating fumes at temperatures up to 200°F.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Lot.- For purposes of sampling, a lot shall consist of all cement from one production batch offered for delivery at one time.

4.2 Sampling.- The Government inspector shall select a representative 1-gallon sample from each lot. The sample so selected shall be forwarded to a Government laboratory designated by the bureau or agency concerned for the tests specified in 4.4. Samples shall be carefully identified as to manufacturer's name, contract number, and lot number.

4.3 Rejection.- If the sample selected in accordance with 4.2 fails any of the tests specified in 4.4, the entire lot shall be rejected. A rejected lot may be resubmitted for Government inspection, but only after the manufacturer, subsequent to the first Government inspection, has reworked the entire lot; subjected the lot to the test or tests wherein failure occurred, and has thereby removed all nonconforming material.

4.4 Preparation of test specimens and testing.-

4.4.1 Fibrous insulating compound conforming to Specification 32C14, shall be applied between the fixed flanges on a section of pipe of the size shown on figure 1, so that the compound will be flush with the flanges. After the insulating compound has become suitably dry, the cement under test shall be used to apply to the surface of the compound three strips of type A asbestos cloth, conforming to Specification MIL-C-1207. Each strip shall be 3 inches wide by 17 inches long and shall be applied to encircle the compound, except that the surplus length shall be left free instead of lapped over to make a seam. The cement shall be allowed to dry

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for 72 hours at room temperature. The pipe section shall then be mounted at the uncovered ends on frictionless rollers so that the pipe will be free to turn about its longitudinal axis. Load shall be applied vertically downward on the free end of each strip of cloth, successively, until the strips are separated from the insulating compound.

4.4.2 A steel plate, 12 inches square and 1/8 inch thick, shall be heavily coated on one side with the test cement. After drying for 7 days at 70°F. the plate shall be placed in an oven and heated to a temperature of 200°F. for 2 hours. No objectionable or irritating fumes shall give off.

5. PREPARATION FOR DELIVERY

5.1 Packaging (for domestic and overseas shipment).- The cement shall be furnished in 1-quart, 1 gallon multiple-friction top containers, or in 5 gallon lug-cover steel drums as specified (see 6.1).

5.1.1 One quart and one gallon multiple-friction top containers shall conform to type IV of Specification JAN-P-124.

5.1.2 Five gallon lug-cover steel drums shall conform to type VI of Specification JAN-P-124.

5.2 Packing.-

5.2.1 For domestic shipment.-

5.2.1.1 Four 1 gallon containers or twelve 1 quart containers shall be packed in nailed wood or wirebound box conforming to Specifications JAN-P-106 and JAN-B-107, respectively, or in a style RSC corrugated or solid fiberboard box having a minimum Mullen test of 275 pounds conforming to Specifications LLL-B-631 and LLL-B-636, respectively. No strapping will be required.

5.2.1.2 Five gallon lug-cover steel drums will require no overpacking.

5.3 Marking.- In addition to any special marking required by the contract or order, intermediate and shipping containers shall be marked with the date of manufacture of the product, and in accordance with the Navy Shipment Marking Handbook.

6. NOTES

6.1 Ordering data.- Requests, requisitions, schedules and contracts or orders should specify the following:

Title, number, and date of this specification.
Size of containers required (see 5.1).
Whether the subject commodity is to be prepared for domestic or overseas shipment (see section 5).

6.2 Fibrous insulation adhesive cement should not be applied directly on any painted metal surface, for the reason that certain compositions of this cement are chemically active on paint.

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6.3 Copies of this specification may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Center, Norfolk 11, Va. Both the title and identifying number or symbol should be stipulated when requesting copies.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Interest:
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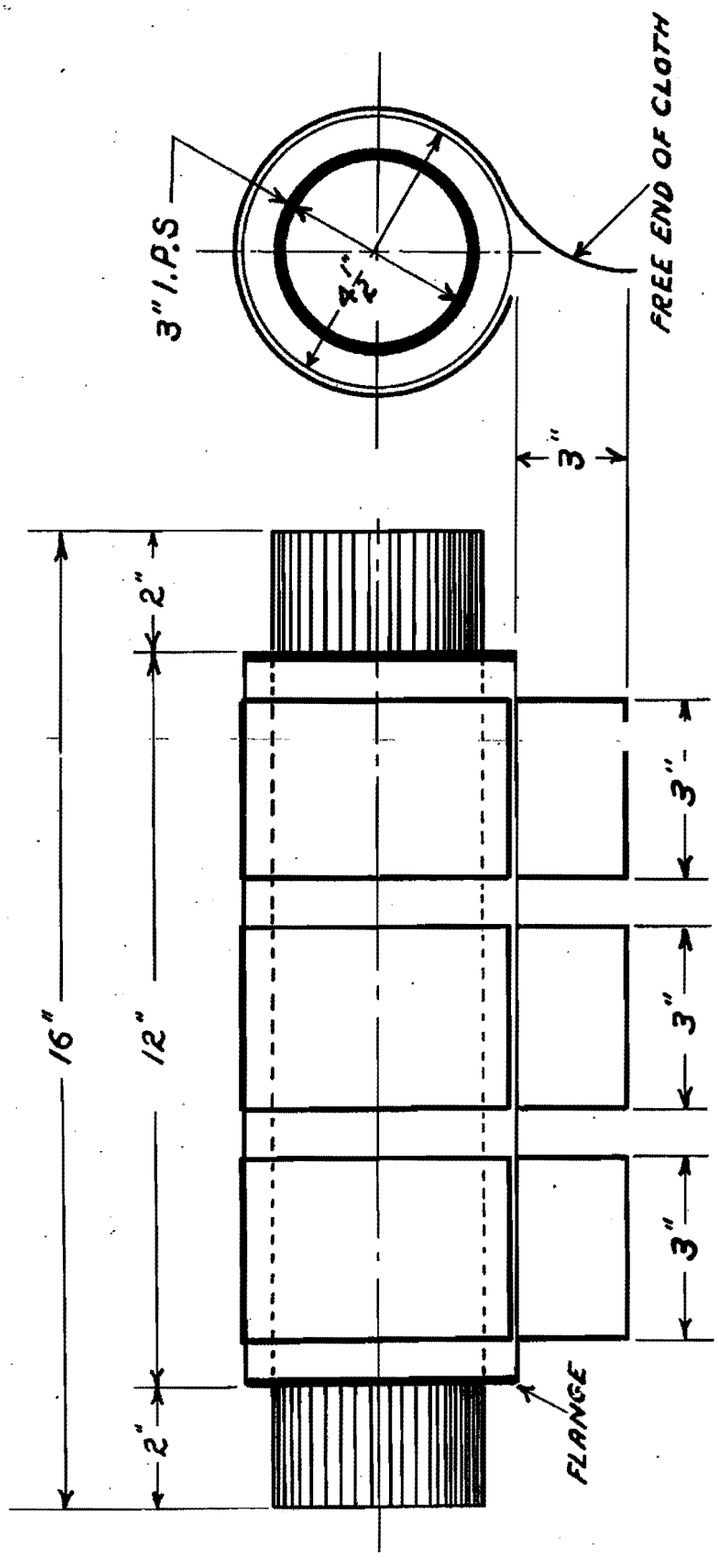


Figure 1. - Adhesive Test Apparatus.

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